Ahmed Taha Bouallegui

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Technical Summary

Computer Engineering student specializing in IoT and Cybersecurity. Linux system administration, infrastructure as code, self-hosted environments built with Nix on NixOS, including a home lab that runs its own models locally. Looking for an internship in systems administration or cybersecurity in summer 2027, 10 weeks to 4 months.

Education

2023 to 2028 **M.Eng. in Internet of Things and Cybersecurity**, *École Polytechnique Universitaire de Lille (Polytech Lille)*, Villeneuve d'Ascq, France

2023 to 2025 **Preparatory Cycle (MPCSI)**, *Polytech Lille*, Villeneuve d'Ascq, France
Graduated with Highest Honors, Top of Class 2023/2024

2023 **Baccalauréat (French high school diploma)**, *France*
Très Bien (Highest Distinction)

Technical Skills

Systems & infra.	Linux (NixOS, Debian, Arch, Ubuntu), KVM/QEMU, Docker, microVMs, systemd.
Declarative config.	Nix/NixOS, sops, reproducible builds, environment management.
Local AI	llama.cpp, vLLM.
Networking & Security	Tailscale, ZeroTier, IAM isolation, AWS (EC2, IAM, security groups), CTF (Reverse Engineering, Web/Memory Safety).
Programming	Python, C/C++, C#, Java, Rust, Bash, PostgreSQL, Assembly (AVR/ARM).
Tools	Git, KiCad, MATLAB/Simulink, Matplotlib.
Languages	English (C1), Arabic (C2), French (C1), German (B1).

Selected Projects

Since Aug 2026 **Local LLM Inference on Consumer Hardware**

- Open-weight models on three machines with `llama.cpp`: two RTX 3060 boxes and one ThinkPad running on CPU only, reachable only through a VPS.
- A 125B mixture-of-experts model on a 12 GB card: experts split across VRAM, RAM and a 29 GB SSD lookup table. 35 tok/s decode and 253 tok/s prefill, against 12 and 55 before the engine change.

Since Jan 2026 **Self-Hosted Linux Server Infrastructure**

- Headless server: Immich, Jellyfin, Grocy, SearXNG and Samba. Started on Debian in January 2026, rebuilt on NixOS in February from one shared config tree.
- Nothing listens on the public internet: services are reachable only through a VPS, and the repository secrets are encrypted with sops.

Jan to Jun 2026 **Split Ergonomic Keyboard (coursework)**
Schematic and two-layer PCB drawn in KiCad, firmware written in C.

Professional Experience

Jun to Jul 2026 **Research Internship, CRIStAL (2XS team)**, *IRCICA* • *Université de Lille*, Lille, France

- Removed every fork and exec of `iw` and `ip` in the wifi module of a Linux IoT-detection probe, replacing them with direct `nl80211` and rtnetlink calls and automatic interface discovery.
- Contributed to the 2XS team's embedded work on memory footprint, energy efficiency and security, and to kernel debugging.

Apr 2025 to now **Fabmanager (Fabricarium FabLab)**, *PolyTech Lille*, Villeneuve d'Ascq, France

- Operate and maintain the school's FabLab: 3D printers, laser cutters, CNC, electronics workstations.
- Support students and researchers prototyping hardware and IoT projects; coordinate bookings, safety and open sessions.

Jul to Aug 2023 **Internship in Automation Engineering (Valeo)**, *Valeo*, Tunis, Tunisia

- Automated a sound-recognition quality check on a steering-wheel line with Python and Librosa.